

Expected Utility and Harsanyi Aggregation

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- ▶ The von Neumann–Morgenstern (VNM) axioms characterize exactly when an agent can be modeled as maximizing expected utility.
- ▶ Harsanyi's theorem characterizes the aggregate expected-utility preferences that satisfy the Pareto property (weighted social welfare).

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Individual input	preference relation	preference relation
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Representation	ordinal utility; only order matters	expected utility; positive-affine scale

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That richer structure permits weighted aggregation—but requires more information than rankings of deterministic completions contain.

A Language-Model Policy Is a Lottery

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- ▶ A completion is an element $j \in C$.
- ▶ A policy is a lottery $p \in \Delta(C)$.

Rankings of Completions Do Not Rank Every Policy

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To evaluate randomized policies, we need preferences defined on all
of $\Delta(C)$.

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- ▶ $p \succsim q$ means that p is weakly preferred to q .
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The first two VNM axioms require $\succeq$ to be a weak order:

1. *Completeness*: for every p, q , either $p \succeq q$ or $q \succeq p$.
2. *Transitivity*: $p \succeq q$ and $q \succeq r$ imply $p \succeq r$.

Continuity Interpolates via Randomization

Definition

The preference satisfies *continuity* if

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implies that, for some $\alpha \in [0, 1]$,

$$q \sim \alpha p + (1 - \alpha)r.$$

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As the probability assigned to the better lottery p rises from zero to one, the resulting mixture cannot pass discontinuously from below q to above q : at some intermediate probability, it must be indifferent to q .

Independence Makes Common Mixtures Cancel

Definition

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Mixing both options with the same third lottery cannot reverse their order.

VNM Characterizes Expected-Utility Maximization

Theorem (von Neumann–Morgenstern)

A preference on $\Delta(C)$ satisfies completeness, transitivity, continuity, and independence if and only if there is a utility vector $u \in \mathbb{R}^C$ such that

$$p \succeq q \iff u^\top p \geq u^\top q.$$

Moreover u is unique up to a positive affine transformation: $\tilde{u}$ represents the same preference exactly when $\tilde{u} = au + b\mathbf{1}$ for some $a > 0$ and $b \in \mathbb{R}$.

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$$u^\top p = \sum_{j \in C} p_j u(j) = \mathbb{E}_{J \sim p}[u(J)].$$

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$$u^\top p = \sum_{j \in C} p_j u(j) = \mathbb{E}_{J \sim p}[u(J)].$$

Thus choosing a most-preferred available lottery is exactly expected-utility maximization. Conversely, every expected-utility preference satisfies the four axioms.

Expected Utility Calibrates Completions Against Lotteries

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A policy $p = (0.2, 0.5, 0.3)$ over (a, b, c) then has expected utility

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The number 0.7 came from an observed indifference. The proof of the theorem turns this calibration trick into a construction.

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If $U_p = U_r$, then all three are equal and we may take $\alpha = 0$. Otherwise, $U_p > U_r$. To make the mixture indifferent to q , its expected utility must satisfy

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Solving,

$$\alpha = \frac{U_q - U_r}{U_p - U_r}.$$

Since $U_p \geq U_q \geq U_r$, this α lies in $[0, 1]$. By construction, the mixture has expected utility $U_q = u^\top q$, so $q \sim \alpha p + (1 - \alpha)r$.

Expected Utility Implies Independence

For every $\alpha \in (0, 1)$,

$$\begin{aligned} u^\top(\alpha p + (1 - \alpha)r) - u^\top(\alpha q + (1 - \alpha)r) \\ = \alpha u^\top(p - q). \end{aligned}$$

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Because $\alpha > 0$, the mixed-lottery difference has the same sign as the original utility difference. Therefore

$$p \succeq q \iff \alpha p + (1 - \alpha)r \succeq \alpha q + (1 - \alpha)r.$$

The VNM Axioms $\implies$ Expected Utility

Now assume that $\succeq$ satisfies the four VNM axioms. We will construct a representing utility vector u :

1. Record two consequences of independence: it preserves strict preference, and it lets us substitute indifferent components inside a mixture.

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4. Use continuity to match every pure completion to a unique best–worst lottery, then substitute to reduce any policy to one.

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4. Use continuity to match every pure completion to a unique best–worst lottery, then substitute to reduce any policy to one.
5. Read off an expected-utility representation and prove its uniqueness.

Independence Preserves Strict Preference and Indifference

Strict preference. Suppose $p \succ q$ and $\alpha \in (0, 1)$. Independence gives $\alpha p + (1 - \alpha)r \succeq \alpha q + (1 - \alpha)r$. If the reverse weak comparison also held, independence applied with p and q interchanged would give $q \succeq p$, contradicting $p \succ q$. Hence

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Indifference. Suppose $s \sim s'$, that is, $s \succeq s'$ and $s' \succeq s$. For $\lambda \in (0, 1)$, applying independence to each comparison gives

$$\lambda s + (1 - \lambda)r \succeq \lambda s' + (1 - \lambda)r \quad \text{and} \quad \lambda s' + (1 - \lambda)r \succeq \lambda s + (1 - \lambda)r,$$

so the two mixtures are indifferent.

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Iterating over a finite support, we may replace *every* component at once: if $s_j \sim s'_j$ for each j , then for any weights $p_j \geq 0$ summing to one,

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This substitution rule is the workhorse of the construction. We use it twice: once immediately, and once at the end.

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Define the best–worst lotteries

$$\ell(\alpha) := \alpha e_{j^+} + (1 - \alpha) e_{j^-}, \quad \alpha \in [0, 1].$$

Any Positive Probability of the Best Beats the Worst

Fix $\alpha \in (0, 1)$. Applying strict-preference preservation to $e_{j+} \succ e_{j-}$, mixing both sides with e_{j-} at weight $1 - \alpha$, gives

$$\ell(\alpha) = \alpha e_{j+} + (1 - \alpha)e_{j-} \succ \alpha e_{j-} + (1 - \alpha)e_{j-} = e_{j-}.$$

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When $\alpha = 1$ this is just $e_{j+} \succ e_{j-}$, and when $\alpha = 0$ both sides coincide.

So $\ell(\alpha) \succ \ell(0)$ for every $\alpha > 0$. Next: compare two *arbitrary* mixing probabilities.

More Weight on the Best Completion Is Strictly Better

Fix $0 \leq \gamma < \alpha \leq 1$; we show $\ell(\alpha) \succ \ell(\gamma)$. If $\gamma = 0$ this is the previous slide, so assume $\gamma > 0$ and set $\lambda = \gamma/\alpha \in (0, 1)$.

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Now apply strict-preference preservation to $\ell(\alpha) \succ e_{j-}$, mixing both sides with $\ell(\alpha)$ at weight λ :

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The parameter α strictly orders the best–worst lotteries.

Continuity Matches Each Completion to a Lottery

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The continuity axiom applies directly to these three lotteries. For some $\alpha_j \in [0, 1]$,

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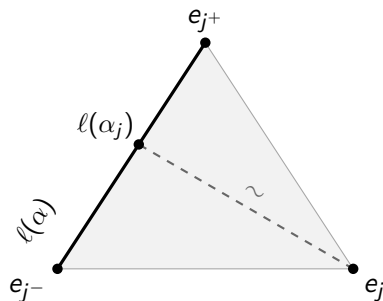
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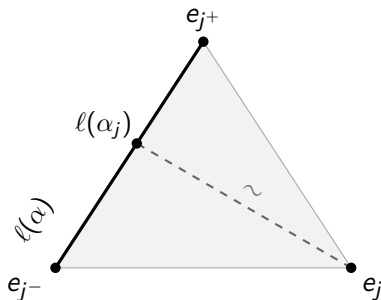
This mixing probability will become the utility of completion j .

The Construction in One Picture



The thick edge is the family $\ell(\alpha)$, strictly increasing in α . Continuity slides a point along it until it is indifferent to e_j ; that point's parameter is α_j .

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Substitution now collapses *any* lottery onto this one edge.

Every Lottery Is Indifferent to One Best–Worst Lottery

Apply the substitution rule with $s_j = e_j$ and $s'_j = \ell(\alpha_j)$:

$$\begin{aligned} p = \sum_j p_j e_j &\sim \sum_j p_j \ell(\alpha_j) \\ &= \ell \left(\sum_j p_j \alpha_j \right), \end{aligned}$$

where the last equality is a direct calculation: a mixture of best–worst lotteries is the best–worst lottery at the mixed parameter.

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Every lottery therefore sits at a single point of the thick edge.

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This proves the expected-utility representation.

The Scale Is Cardinal but Not Interpersonal

Our construction normalized

$$u(j^+) = 1, \quad u(j^-) = 0.$$

Under this normalization, each indifference $e_j \sim \ell(\alpha_j)$ forces $u(j) = \alpha_j$.

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Ratios of utility differences now have meaning within one person's scale. Absolute levels and units do not, and the theorem does not equate one person's utility unit with another person's.

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We now fix a profile $u_1, \dots, u_n$ — one representation per person — and ask which aggregate preferences respect unanimity.

Aggregation Must Also Output a Preference Over Lotteries

Fix n people with VNM preferences represented by

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At this fixed preference profile, an aggregation procedure must output some preference over lotteries. Call a possible output a *candidate aggregate preference*. Require it to satisfy the VNM axioms, and let u_0 be one of its representations.

Weak Pareto Compares Unanimous Lottery Improvements

Definition

The candidate aggregate satisfies the *weak Pareto axiom* if, for every $p, q \in \Delta(C)$,

$$u_i^\top p \geq u_i^\top q \quad \text{for every } i \quad \implies \quad u_0^\top p \geq u_0^\top q.$$

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Whenever every person weakly prefers p to q , the aggregate may not rank q strictly above p . The axiom does not specify what happens when people disagree.

Nonnegative Weighted Sums Work

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If every person weakly prefers p to q , then

$$\begin{aligned} u_0^\top(p - q) &= c\mathbf{1}^\top(p - q) + \sum_i w_i u_i^\top(p - q) \\ &= \sum_i w_i u_i^\top(p - q) \geq 0, \end{aligned}$$

because $\mathbf{1}^\top p = \mathbf{1}^\top q = 1$.

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because $\mathbf{1}^\top p = \mathbf{1}^\top q = 1$. Thus such aggregates exist—for example, choose every $w_i = 1$.

Harsanyi's Aggregation Theorem

Theorem (Harsanyi)

Fix individual VNM preferences over $\Delta(C)$, represented by $u_1, \dots, u_n$. A candidate aggregate VNM preference represented by u_0 satisfies the weak Pareto axiom if and only if

$$u_0 = c\mathbf{1} + \sum_{i=1}^n w_i u_i$$

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The easy direction showed that every nonnegative weighted sum works. The converse says that weak Pareto permits no other VNM aggregate.

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3. Form an LP that searches for a direction z that benefits every person but harms the aggregate.
4. Read the dual solution as nonnegative welfare weights plus one additive constant.

Lottery Differences Form a Vector Space

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Conversely, every $z \in V$ is proportional to a lottery difference. Let $\bar{p} = \mathbf{1}/|C|$. For sufficiently small $t > 0$,

$$p := \bar{p} + tz \in \Delta(C), \quad q := \bar{p} \in \Delta(C),$$

and therefore $p - q = tz$.

Weak Pareto Becomes a Sign Implication

Fix $z \in V$ with $u_i^\top z \geq 0$ for every person $i = 1, \dots, n$, and use the lotteries satisfying $p - q = tz$ from the previous slide.

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$$0 \leq u_0^\top (p - q) = t u_0^\top z.$$

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Every statement about lotteries has now become a statement about signs of linear functions on V .

One Sign Implication Must Force the Weighted-Sum Form

We have converted the weak Pareto axiom into the implication

$$u_i^\top z \geq 0 \quad \forall i \quad \implies \quad u_0^\top z \geq 0$$

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An LP will search for a violation of the implication; its dual will instead produce the coefficients $w_1, \dots, w_n$ and c .

LP Duality Will Supply the Weights

For a primal variable z whose coordinates may have either sign,

$$(P) \quad \max_z \quad g^\top z \quad \text{subject to } Gz \leq h, \quad Ez = f,$$

$$(D) \quad \min_{w, \lambda} \quad h^\top w + f^\top \lambda \quad \text{subject to } G^\top w + E^\top \lambda = g.$$

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Watch what the dual constraint will say: it is an *equation expressing* u_0 in terms of the other utility vectors.

The Primal Searches for a Pareto Violation

Collect the individual utility vectors as

$$U = [u_1 \ \cdots \ u_n], \quad U^\top z = (u_1^\top z, \dots, u_n^\top z)^\top.$$

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The Equality Multiplier Is the Constant

Write $U^\top z \geq 0$ as $-U^\top z \leq 0$. Give it a multiplier $w \geq 0$, and give $\mathbf{1}^\top z = 0$ an unrestricted multiplier γ . The dual is

$$\begin{array}{ll} \min & 0 \\ & w \geq 0, \gamma \in \mathbb{R} \\ \text{subject to} & -Uw + \gamma \mathbf{1} = -u_0. \end{array}$$

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The primal is feasible and bounded above, so LP duality says that the dual has a feasible solution. Rearranging its equality and writing $c = -\gamma$ gives

$$u_0 = Uw + c\mathbf{1} = c\mathbf{1} + \sum_{i=1}^n w_i u_i, \quad w_i \geq 0.$$

The Converse Completes the Theorem

Before stating the theorem, we proved the easy direction:

$$u_0 = c\mathbf{1} + \sum_i w_i u_i \text{ with } w_i \geq 0 \implies \text{weak Pareto.}$$

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Together, the two implications prove Harsanyi's characterization. $\square$

Harsanyi Does Not Choose the Weights

Harsanyi characterizes the form of every weak-Pareto VNM aggregate:

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The same observed preferences support different numerical weights. Interpersonal utility scales are not identified from data.

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- ▶ Once interpersonal utility scales are chosen, Harsanyi characterizes the weak-Pareto aggregates as nonnegative weighted sums.
- ▶ Choosing the scales and welfare weights is an additional normative decision; it is not determined by observed preferences alone.

Next time: RLHF observes neither the utilities u_i nor the weights w_i — only which of two completions a labeler preferred. We ask which aggregate a scalar reward model builds out of that far poorer data.

Thanks!

See you next class!

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